

- INTERLOCKING PROPERTIES
1. SECTION COLLISION PREVENTION
Every physical track section begins with one token in its Free place.

A train entering a section consumes that token and restores it when leaving.
 2. CROSSING COLLISION PREVENTION
Passenger 1 -> 5 requires Upper Crossing.
Passenger 6 -> 2 requires Lower Crossing.
Freight 3 <-> 4 requires BOTH resources.
 3. PASSENGER PRIORITY
Freight 3 <-> 4 additionally requires
Section 1 Free and Section 6 Free.

If a passenger occupies either crossover
approach section, freight movement is disabled.
 4. DIRECTION PRESERVATION
Northbound and southbound occupied states are
represented separately.
 5. PASSENGER / FREIGHT SEPARATION
No passenger place has a transition into a
freight state and vice versa.
 6. FREIGHT DEADLOCK PREVENTION
The workshop and main freight routes use
reservation tokens.
 7. VALID ROUTES

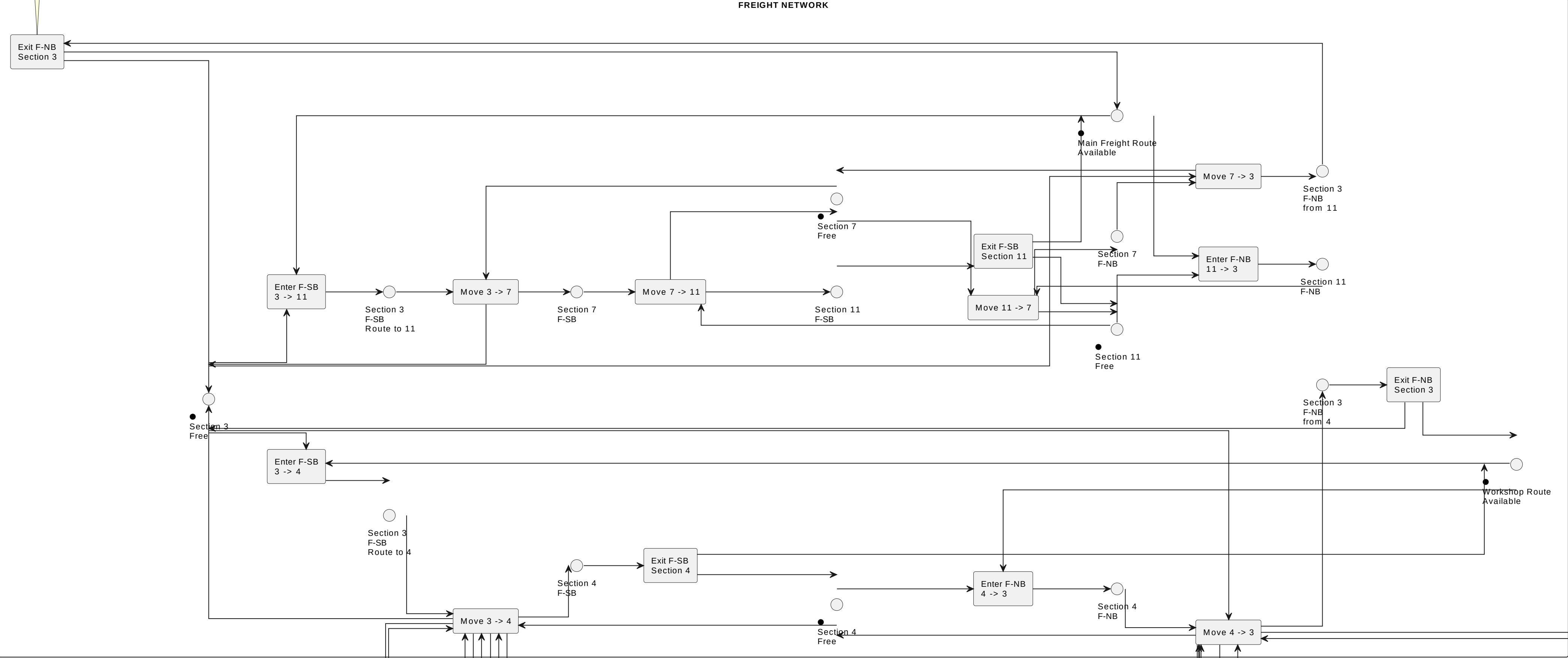
Passenger Southbound:
1 -> 5 -> 8
1 -> 5 -> 9

Passenger Northbound:
9 -> 6 -> 2
10 -> 6 -> 2

Freight Southbound:
3 -> 4
3 -> 7 -> 11

Freight Northbound:
4 -> 3
11 -> 7 -> 3

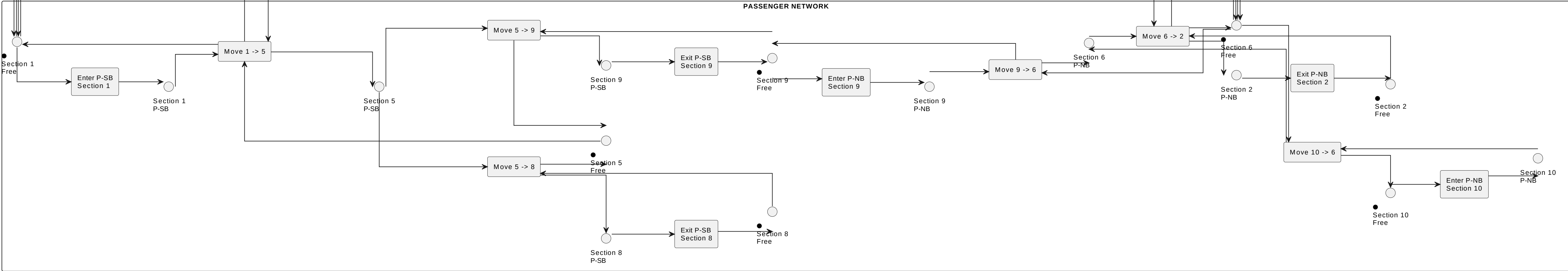
FREIGHT NETWORK



FREIGHT / PASSENGER CROSSOVER

- CROSSOVER CONTROL
- Passenger movement 1 -> 5:
requires Upper Crossing.
- Passenger movement 6 -> 2:
requires Lower Crossing.
- Freight movement 3 <-> 4:
requires BOTH crossings.
- PASSENGER PRIORITY
- Freight movement additionally requires:
Section 1 Free
Section 6 Free
- These places are connected to the freight
transition by self-loops.
- Therefore, if a passenger is waiting in
Section 1 or Section 6, the corresponding
Free token is absent and freight movement
across the passenger lines is disabled.

PASSENGER NETWORK



Symbol	Meaning
●	Initial token
P-SB	Passenger Southbound
P-NB	Passenger Northbound
F-SB	Freight Southbound
F-NB	Freight Northbound

Initial marking M0:
S1_Free, S2_Free, S3_Free, S4_Free,
S5_Free, S6_Free, S7_Free, S8_Free,
S9_Free, S10_Free, S11_Free,
CrossUpper, CrossLower,
WorkshopAvailable, MainAvailable.